

Mahmoud Ahmed Abdelawal

AI & Data Science Engineer | Computer Science Graduate

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[LinkedIn](#) | [GitHub](#) | [Kaggle](#)

PROFESSIONAL SUMMARY

AI and Data Science Engineer specializing in machine learning, deep learning, computer vision, and natural language processing. Experienced in building multimodal AI systems, generative AI applications, disaster response models, and production-ready FastAPI backends. Skilled in Python, PyTorch, TensorFlow, Scikit-Learn, Docker, Azure, and optimized AI inference workflows.

TECHNICAL SKILLS

AI & Machine Learning: Machine Learning, Deep Learning, Computer Vision, NLP, Generative AI

Computer Vision: YOLOv8, Segmentation, Image Classification, OpenCV, MiDaS Depth Estimation, CNNs

NLP & RAG: Text Preprocessing, Tokenization, Embeddings, Sequence Modeling, ChromaDB, Vector Search

Frameworks & Libraries: PyTorch, TensorFlow, Scikit-Learn, CLIP, BLIP, ConvNeXt, GANs, Gemini API

Programming Languages: Python, Java, C++, JavaScript, TypeScript

Backend & DevOps: FastAPI, React, SQLite, Git, GitHub, Microsoft Azure, Docker, VS Code

EDUCATION

Bachelor of Computer Science and Information Technology | Egyptian E-Learning University (EELU), Sohag

Specialization: Artificial Intelligence | Graduation: June 2026

PROJECTS

S-Report AI: Universal Urban Intelligence & Disaster Management Graduation Project

■ <https://github.com/MahmoudMahrous24/S-Report-AI-Vision>

- Developed a hybrid multimodal AI system using CLIP zero-shot learning and ConvNeXt-Tiny for disaster classification.
- Integrated BLIP, YOLOv8 segmentation, and MiDaS depth estimation for spatial intelligence and scene understanding.
- Engineered a weighted decision and dispatch engine for automated emergency resource allocation.
- Architected a production-ready FastAPI backend with SQLite and Docker containerization.

DermaAssist AI — Clinical Decision Support System ■ 1st Place · AI Hackathon

■ <https://github.com/MahmoudMahrous24/AI-Hackathon-Sohag>

- Built a bilingual (Arabic/English) AI dermatology platform grounded in official WHO clinical guidelines.
- Implemented RAG pipeline using ChromaDB + Nomic Embeddings + Gemini Flash for evidence-based responses.
- Developed multimodal skin lesion analysis (image/video), voice query support (Arabic & English), and a Doctor/Patient portal with prescription management.
- Stack: React 19 + TypeScript (frontend) · FastAPI + SQLite + SentenceTransformers (backend).
- Won 1st place at AI Hackathon sponsored by Ministry of Communications (Egypt), Orange, and Instant.

Artificial Vincent Van Gogh DEPI Project

■ <https://github.com/DEPI-Project/Artificial-Vincent-Van-Gogh>

- Built a generative AI system using DCGANs and neural style transfer to generate Van Gogh-inspired images.
- Trained deep learning models on a curated Van Gogh dataset using Python and GAN-based workflows.

Data Analysis Projects Data Analysis

■ <https://github.com/MahmoudMahrous24/Data-Analysis>

- Conducted EDA, data cleaning, visualization, and insight extraction across real-world datasets.
- Built analytical pipelines using Python, Pandas, NumPy, Matplotlib, and Seaborn.

CERTIFICATIONS & TRAINING

Microsoft ML Engineer — Digital Egypt Pioneers Initiative (DEPI) (12/2025)

English and Soft Skills — Digital Egypt Pioneers Initiative (DEPI) (2025)

Deep Learning, Computer Vision, and NLP — Hisham Asem (2024)

ACHIEVEMENTS

★ **Ministry Recognition:** Honored by the **Minister of Communications of Egypt** for outstanding projects and recognized as one of the top-performing students nationwide within DEPI.

■ **1st Place — AI Hackathon:** Won first place with DermaAssist AI, a bilingual clinical decision support system for dermatology. Hackathon sponsored by the **Ministry of Communications (Egypt)**, **Orange Egypt**, and **Instant**.